

Thomas John Neuenfeldt

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EDUCATION

University of Michigan

Bachelor of Science Engineering in Computer Science; Minor in Music

GPA: 3.96 / 4.0

Coursework: Computational Linear Algebra, Discrete Mathematics, Algorithms and Data Structures, Human-Robot Interaction, Computer Organization, Machine Learning

Ann Arbor, MI

May 2026

SKILLS

Backend: Java, C/C++, Python, Django, Julia, MATLAB/Simulink, PyTorch

Frontend: Vue.js, HTML, JavaScript, CSS

Computer: ROS, Linux/Unix systems, Git

WORK EXPERIENCE

GE Vernova

Onshore Wind Fleet Operations Intern

Schenectady, NY

May 2024 – Present

- Optimize proprietary wind turbine software to improve efficiencies and revenue by analyzing over 4,000 specialized instructions and 5,000 configuration changes to determine their validity for removal or extension.
- Identify and prioritize which fault handling procedures should be automated, adhering to the goal of 70% increase in procedure automation by year-end, thus reducing the 7,500 daily customer service cases and reducing manpower.

Nexteer Automotive

Software Development Intern

Saginaw, MI

May 2023 – August 2023

- Collaborated with senior software engineers to design and program a customer requirement intake tool which identifies requirement changes based on a Simulink model using Python, MATLAB/Simulink, and HTML.
- Transformed existing OEM requirements command-based system to a user-friendly interface using MATLAB/Simulink.
- Strengthened intern collaboration through designing, constructing, and racing a Go-Kart using recycled materials from the plant.

Michigan State University

Material Science Research Intern

Midland, MI

June 2022 – August 2022

- Analyzed and prioritized various biodegradable plastic materials using 3D printers, Autodesk Fusion360, and nTopology to identify the ideal structures which will facilitate wound healing and replace traditional metal rods utilized in surgeries.
- Authored a formal research paper utilizing doctorate research methodology and presented to the American Chemistry Society.

PROJECT EXPERIENCE

University of Michigan MROver Project Team

Teleoperations Lead

Ann Arbor, MI

August 2022 – Present

- Lead a team in designing and implementing a user-friendly interface utilizing Vue.js and Django to enable bidirectional communication with the rover, which operates on a ROS/Linux system.
- Accountable for team cohesiveness and sustainability including recruiting, onboarding, and knowledge transfer with the help of the MROver GitHub repository.

University of Michigan

Traveling Salesperson Solver

Ann Arbor, MI

August 2023 – December 2023

- Implemented a branch-and-bound solver using C++ to find the most efficient path that visits each node only once in a graph. Tested different greedy heuristics to find the close-to-optimal path taken.

Image Classification

January 2024 – April 2024

- Using PyTorch (Python), developed a custom convolutional neural network model to classify landmarks across Europe and applied transfer learning to the model to improve training.

FIRST Robotics

Lead Programmer, Student Director

Midland, MI

September 2012 – May 2022

- Designed and programmed a robot using Java to autonomously complete tasks. Utilized SCRUM methodology to optimize project management milestones. Enforced gracious professionalism amongst team members.

ACTIVITIES

University of Michigan Marching Band, Member, Rank Leader

August 2022 – Present

4-H Community Service, Historian, Volunteer

September 2008 – June 2022